



TRAINING EXAMPLE — NOT A REAL IEC 62304 DELIVERABLE

This project is a **training site, not a real medical device** under any regulatory definition. The device described in this document — “SentinelFlow 500” — is entirely **fictional**, invented for this course: nothing here is a genuine IEC 62304 deliverable, a real regulatory submission, or evidence of any real organisation’s compliance with anything. It illustrates the *kind* of document a real SaMD/SiMD project would produce. See the [document register](#) for the full explanation.

Clause: 5.7 — Software System Testing · **Register:** [Device Example Register](#)

Document ID: SOP-08 · **Device:** SentinelFlow 500 (fictional) · **Safety class:** C

Status: SOP — template (Procedure not yet written for SentinelFlow 500)

Fictional worked example. See [SOP-01](#) for the disclaimer that applies to this whole document set, and the [Device Example Register](#) for the SentinelFlow 500 device description and what “template” status means here.

1. Purpose

This SOP defines how SentinelFlow 500's complete software system is tested against every documented software requirement, per Clause 5.7.

2. Scope

Applies to system testing of SentinelFlow 500's control software (Class C — see [SOP-01 §7.3](#)). Covers Clause 5.7 in full. Every sub-clause of 5.7 applies to all three safety classes — Amendment 1 moved §5.7.1–5.7.4 from Class B/C to all classes, and §5.7.5 is treated as all-classes on the grounds recorded in

`applicability.json`'s `_resolvedItems` (see the site's Learn page for that reasoning in full).

3. Responsibilities

ROLE	RESPONSIBILITY UNDER THIS SOP
Test Lead	Establishes the system test set; executes tests and records results
Software Development Lead	Maintains traceability between requirements and tests
QA/RA	Evaluates test results against pass/fail criteria; confirms retesting after changes

4. Definitions & Abbreviations

TERM	MEANING
System testing	Testing the complete, integrated software system against its documented requirements
Traceability	The recorded link from a software requirement to the test(s) that verify it

5. References

- IEC 62304:2006+AMD1:2015, §5.7
- **SOP-03 — Software Requirements Specification:** the requirements this test set is written against
- **SOP-13 — Software Problem Resolution Process:** anomalies found during system testing are managed through this process

6. What the standard requires

REF	TITLE	APPLIES AT CLASS C	WHAT MUST BE PRODUCED
5.7.1	Establish tests for software requirements	Yes	A set of tests covering all software requirements, expressed as input stimuli, expected outcomes, pass/fail criteria and procedures
5.7.2	Use software problem resolution process	Yes	(No standard-named output.)
5.7.3	Retest after changes	Yes	(No standard-named output.)
5.7.4	Evaluate software system testing	Yes	Recorded traceability between software requirements and the tests or other verification, plus evidence that results meet the pass/fail criteria
5.7.5	Software system test record contents	Yes	Software system test records: reference to test case procedures, the result and anomalies, the software version tested, the hardware and software test configuration, test tools, date, and the identity of the person responsible

(Quoted directly from `applicability.json` , identical in content to the self-referential set's [08](#), whose Status is itself marked "Documented — strong evidence".)

7. Procedure

Not yet written for SentinelFlow 500 — see the [Device Example Register](#) for what "template" status means. Once complete, this section would give SentinelFlow 500's actual system test set — e.g. dose-rate boundary tests, occlusion-alarm response-time tests, free-flow prevention tests during a simulated set change — with pass/fail criteria and full traceability back to the requirements in SOP-03.

5.7.1 — Establish tests for software requirements

[Template.]

5.7.2–5.7.3 — Problem resolution and retesting

[Template — see SOP-13.]

5.7.4 — Evaluate software system testing (traceability)

[Template.]

5.7.5 — Test record contents

[Template.]

8. Records

RECORD	PRODUCED BY	RETAINED IN
System test set (procedures, expected results)	Test Lead	Project technical file
System test records, per requirement	Test Lead	Project technical file
Requirements-to-test traceability matrix	Software Development Lead	Project technical file

9. Revision History

Illustrative only — SentinelFlow 500 has no real revision history.

VERSION	STATUS	DESCRIPTION
1.0	Template	Structure and requirements table complete; Procedure section pending

